

Mean Fragility and Apparent Structure

Student Worksheet: reading structure off a graph without being fooled

Name: _____

Class / Set: _____

Date: _____

Companion essay read? ☐ Yes ☐ No

What you are doing today. A graph is about to show you something that looks like a deep result, and your job is to find out whether it really is one before you let yourself believe it. You will work inside the interactive tool at https://leeleamacjames.github.io/pascal_diagonal_means_learner.html, which plots the ratio of consecutive averages, $R_n = M_{n+1}/M_n$, for the rows or diagonals of five combinatorial triangles.

The python code can be found at: [googlecolabpythonnotebook](#). The tool gives you four controls, and each one corresponds to a step in a fixed sequence of reasoning that good quantitative thinkers use whenever a pattern appears in data. You will climb that sequence one rung at a time, writing down at every stage both what you see and what the picture is tempting you to claim.

The four controls, and what each is really for.

- **Triangle family** (Pascal, Catalan, Motzkin, Delannoy, Worpitzky): changes the *object* you are measuring.
- **Mean type** (Arithmetic A , RMS Q , Geometric G , Harmonic H): changes the *instrument* you measure with.
- **Chart mode** (raw R_n versus normalised R_n/n): strips out plain growth so any leftover wobble stands out.
- **Row / highlight sliders**: change how much data you look at and where.

The ladder of inference (keep this visible the whole time). 1. **Perceive** what the graph shows. 2. **Name** the theory it tempts you toward. 3. **Vary the instrument**. 4. **Vary the object**. 5. **Find the mechanism**. 6. **State only what is now earned**.

Stage 1 Perceive: set Pascal + Arithmetic, raw ratio

Select **Pascal Fibonacci diagonals**, mean **Arithmetic A** , chart mode **raw R_n** . Set the maximum row reasonably high (say 40+) so the late behaviour is visible.

A trained eye fuses three different things into one impression. Force them apart. Describe, separately and in your own words:

(a) The envelope (how the overall scale behaves as n grows):

(b) The oscillation (any short-range up-down pattern between neighbouring n):

(c) The late regime (what the ratio seems to settle toward):

(d) The arithmetic ratios hover near a famous constant. Read it off the graph to 2 d.p.: _____

What constant do you think this is, and why might Fibonacci numbers be involved?

Stage 2 Name the temptation

The shape you have just described, an oscillation riding underneath a growing envelope, appears all over physics. Before going further, write down what the picture is *inviting* you to claim. Naming the temptation out loud is how you get the chance to resist it.

(a) List every interpretation this graph is nudging you toward. Prompts, if you are stuck: a wave packet with a fast *phase* and a slow *group* envelope; a beat pattern; a period-two cycle; the period-doubling cascade that signals Feigenbaum chaos.

(b) Pick the most dramatic of these (period doubling). What would have to be *true of the system* for that label to be honest? Hint: a period-doubling cascade needs a particular ingredient that you can tune.

(c) Does the tool let you tune any such ingredient? If not, what does that already tell you about the period-doubling reading?

Stage 3 Vary the instrument: cycle through the four means

Keep Pascal and raw R_n . Switch the mean through A, Q, G, H in turn. For each, read the ratio at one even n and the adjacent odd n (use the highlight slider; any pair in the high- n region is fine).

Mean	R_n at even n	R_n at odd n	Even–odd gap
Arithmetic A			
RMS Q			
Geometric G			
Harmonic H			

Rule 1: *if a pattern appears under only one mean, suspect the mean before the object.*

(a) Under which mean is the even–odd gap largest? Under which is it smallest? _____

(b) The oscillation does not vanish when you change instrument, but it changes loudness dramatically. What does that mixed result tell you? Is the oscillation purely an artifact of one mean, purely a property of the object, or something in between?

(c) The harmonic mean is built from reciprocals, $H_n = L_n / \sum_i x_i^{-1}$, so it is dominated by the *smallest* numbers in each diagonal. Why might that make it the most sensitive to what happens at the *ends* of a diagonal?

Stage 4 Vary the object: change the triangle family

Now hold the harmonic mean H fixed (it shouted loudest) and change the family. For each, decide whether the persistent even–odd oscillation is present.

Family	Parity oscillation?	One-line description of what you see
Pascal (Fibonacci diag.)	☐ Y ☐ N	
Catalan / ballot	☐ Y ☐ N	
Motzkin path	☐ Y ☐ N	
Delannoy	☐ Y ☐ N	
Worpitzky	☐ Y ☐ N	

Rule 2: *if a pattern survives several means, vary the underlying family before claiming it is intrinsic.* Catalan and Motzkin are close controls (they still count paths); Delannoy is broader; Worpitzky is deliberately distant.

(a) In how many of the five families does the persistent parity oscillation appear? _____

(b) A control is only useful if it shares the suspect feature while differing in the thing you are testing. Why is Catalan a *fairer* test of “is this just a Pascal thing?” than Worpitzky is?

(c) Combine Stages 3 and 4. The oscillation is loud only under some means *and* present only in one family. Write the most accurate one-sentence verdict you can give so far.

Stage 5 Find the mechanism

A pattern that survives this far deserves an explanation, not just a name. Here the explanation is short, and the tool's "triangle display" helps you see it. Set the family back to Pascal and look at the actual diagonal entries.

Guided reasoning. The arithmetic mean of a Pascal diagonal is

$$A_n = \frac{F_{n+1}}{\lfloor n/2 \rfloor + 1}, \quad \text{so} \quad \frac{A_{n+1}}{A_n} = \underbrace{\frac{F_{n+2}}{F_{n+1}}}_{\rightarrow \varphi} \times \underbrace{\frac{\lfloor n/2 \rfloor + 1}{\lfloor (n+1)/2 \rfloor + 1}}_{\text{parity correction}}.$$

(a) The first factor explains the *envelope*. Which constant is it, and which feature of the graph does the second factor explain?

(b) Now the harmonic case. Look at the two end entries of an *even* diagonal and an *odd* diagonal in the triangle display. The even diagonal ends $(1, \dots, 1)$; the odd diagonal ends $(1, \dots, m+1)$. Compute the contribution of just the two end entries to the reciprocal sum $\sum x_i^{-1}$ in each case:

Even diagonal: $1^{-1} + 1^{-1} = \underline{\hspace{2cm}}$ Odd diagonal: $1^{-1} + (m+1)^{-1} \approx \underline{\hspace{2cm}}$ (for large m)

(c) Using your two numbers, explain in words why the harmonic ratio pulses so strongly between even and odd n . Is this a "deep dynamical" effect, or a boundary effect made loud by the choice to average reciprocals?

Stage 6 State only what is now earned

You have climbed the whole ladder. Now write the calibrated conclusion: the most the evidence permits, and no more. A good conclusion will say what is *real*, name what it merely *resembled*, and explain the difference.

(a) Is the even–odd structure real, or an illusion of the mean? Justify in one sentence.

(b) Was the period-doubling / Feigenbaum reading justified? Say precisely what was missing.

(c) Write your final verdict on the Pascal graph in three to four sentences, in the form: *the pattern is ... , it is caused by ... , and it is not ... because ...*

Transfer take the ladder out of mathematics

The point of today was never really the triangles. The same six rungs apply whenever a graph seems to contain a result.

Choose *one* of the following and apply the ladder to it in note form: a news chart showing a worrying “cycle” in monthly figures; a science-fair experiment where a trend appears only when you average a certain way; a sports statistic that seems to show a streak. For your chosen case, write one line per rung: what is perceived, what theory it tempts, how you would vary the instrument, how you would vary the object, what mechanism you would look for, and what you could honestly conclude.

First see the structure, then try to destroy it. If it survives, it may deserve a theorem.